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Prof. Daniel Kirchen
Editor-in-Chief
IEEE Transactions on Markets, Policy and Regulation

Dear Prof. Kirchen,

We are pleased to submit our manuscript titled "**Learning-based optimisation framework for flexibility aggregators in ancillary service markets**" for consideration in the IEEE Transactions on Markets, Policy and Regulation.

This work addresses the challenge of optimizing offering strategies for flexibility aggregators operating in the European manual frequency restoration reserve (mFRR) market. Our research proposes a constraint learning-based optimization framework that integrates machine learning surrogate models into optimization problems. Specifically, we introduce a novel linear programming (LP) reformulation for rectified linear unit (ReLU) deep neural networks (DNNs), enabling scalable and accurate modelling of prosumers' price responsiveness and flexibility behaviour.

Key contributions of this paper include:

1. A tight LP reformulation of ReLU DNNs for computational efficiency and scalability in optimization frameworks.
2. Application of the proposed approach to a profit maximization problem for a behind-the-meter (BTM) flexibility aggregator in the mFRR capacity market.
3. Demonstration of superior performance in runtime and solution quality compared to existing methods through extensive simulations.

We believe this work aligns closely with the journal's focus on market frameworks, pricing strategies, and regulatory policies, particularly in enabling flexible energy markets and integrating demand-side resources into ancillary services markets.

We confirm that this manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have approved the manuscript and agree with its submission to the IEEE Transactions on Markets, Policy and Regulation.

Thank you for considering our submission. We look forward to your review and feedback.

Sincerely,

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